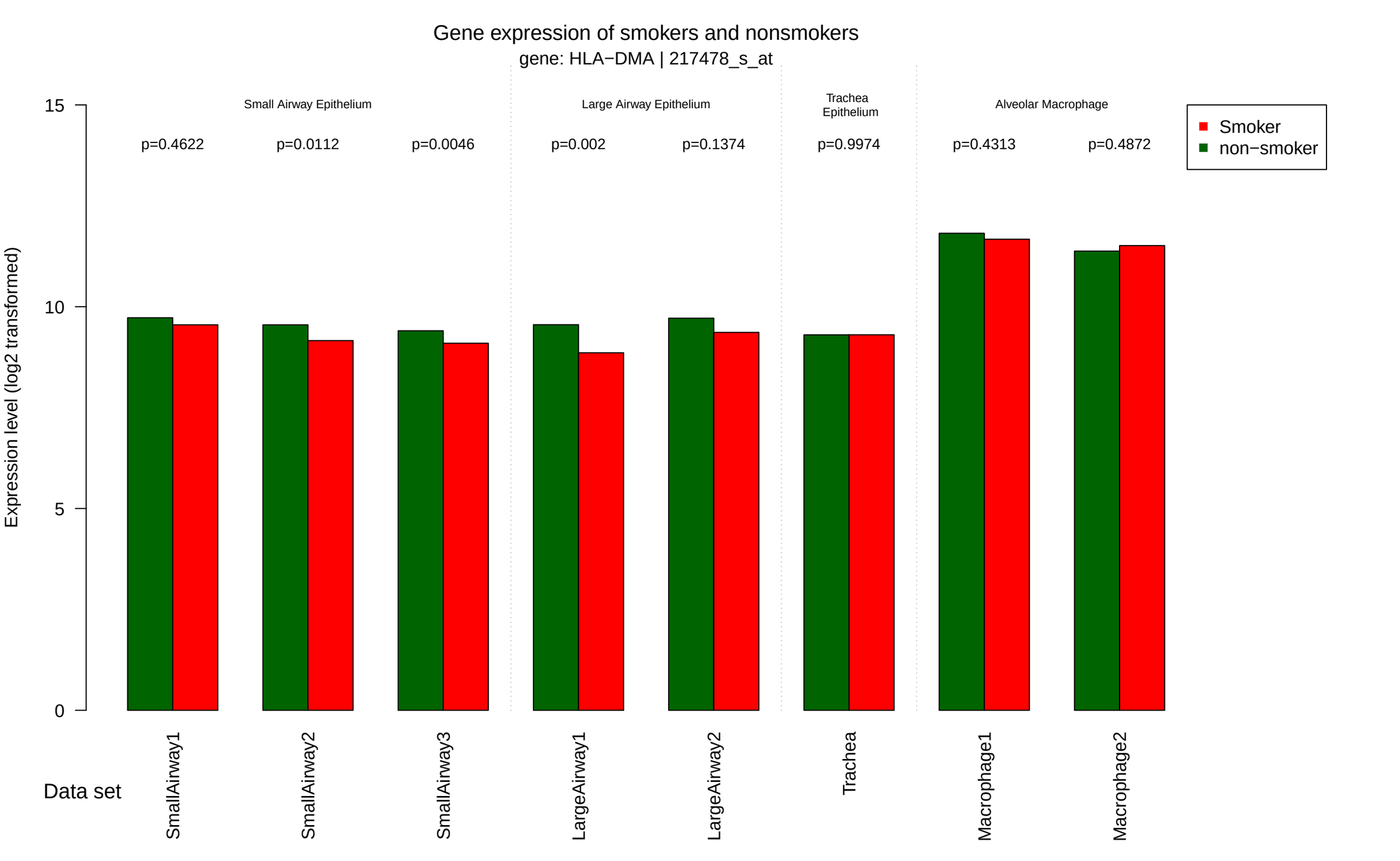
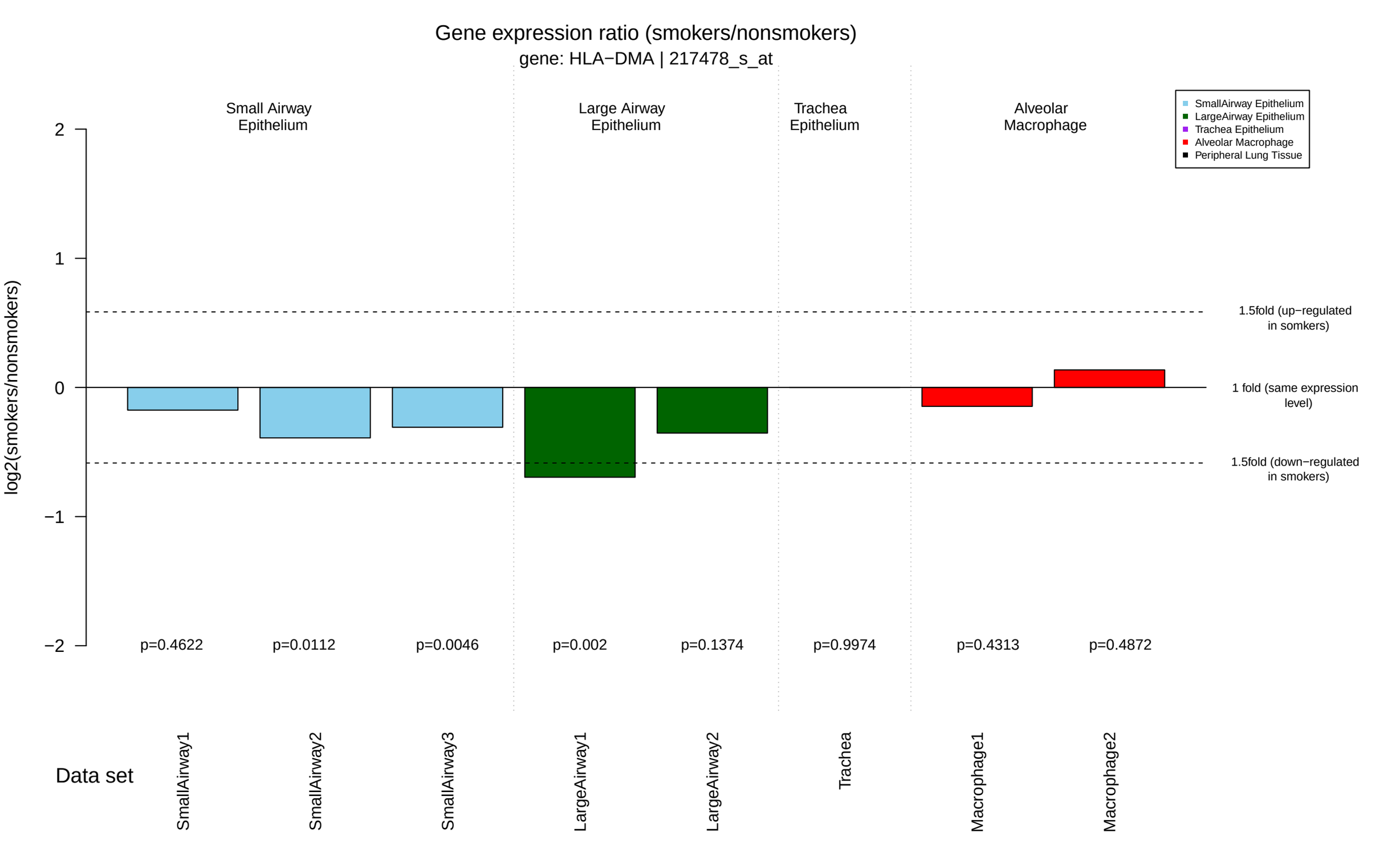
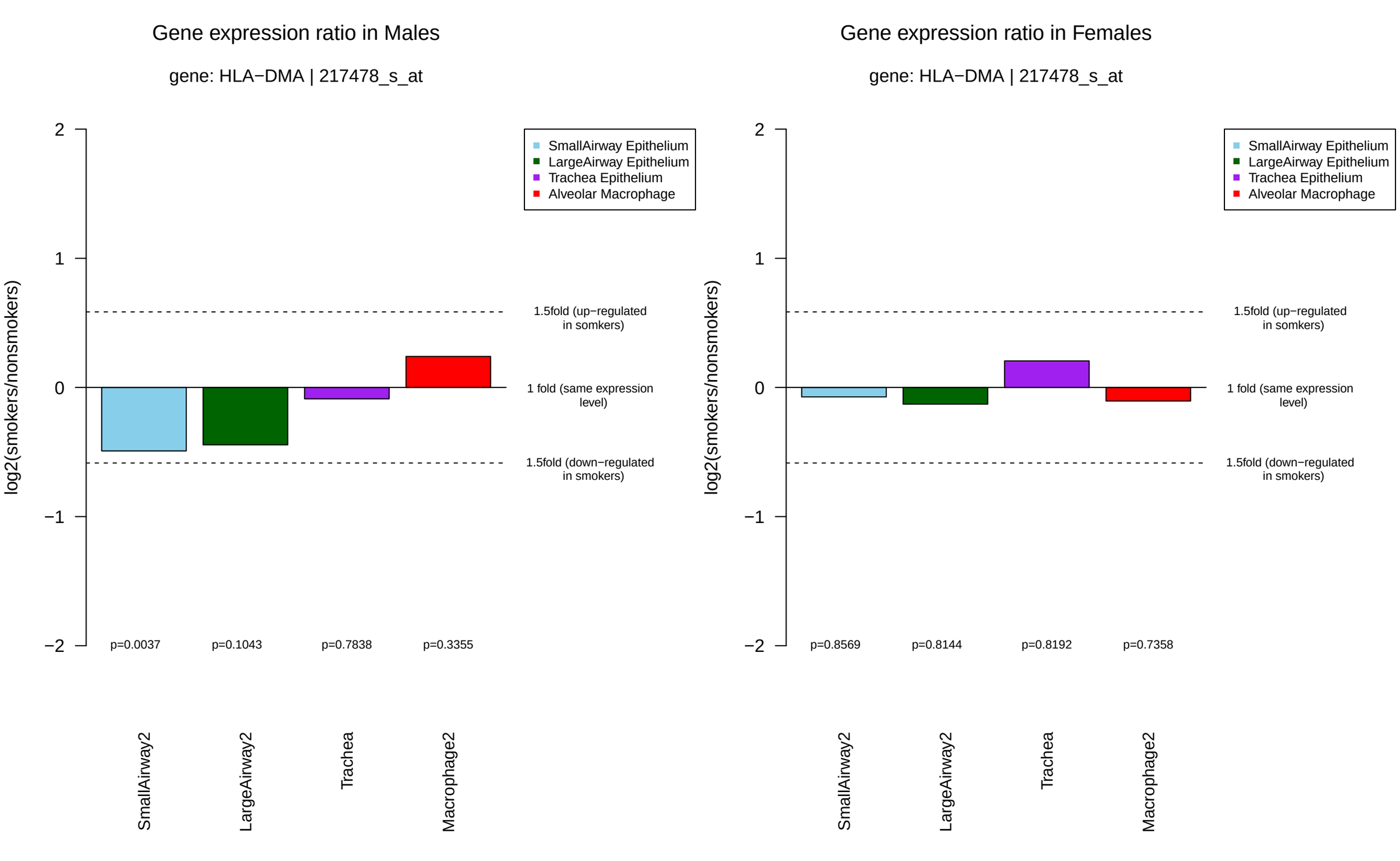
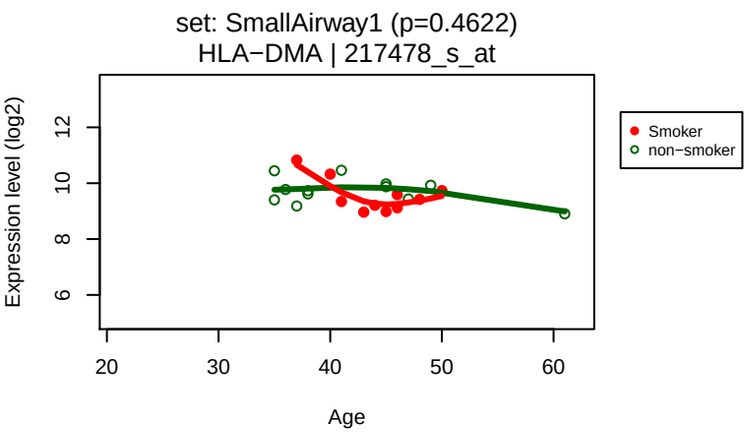
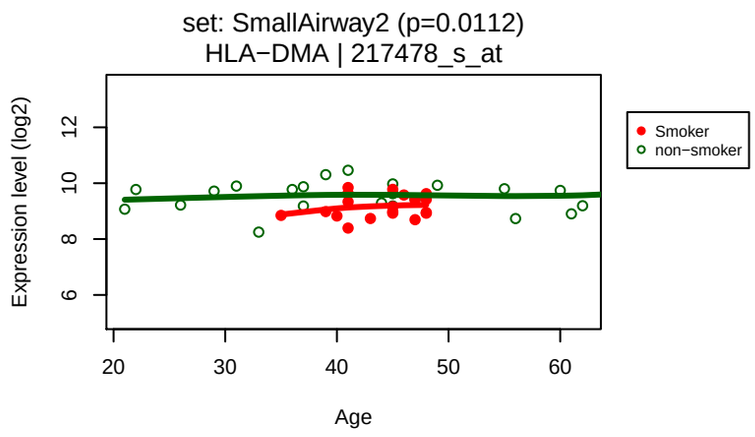
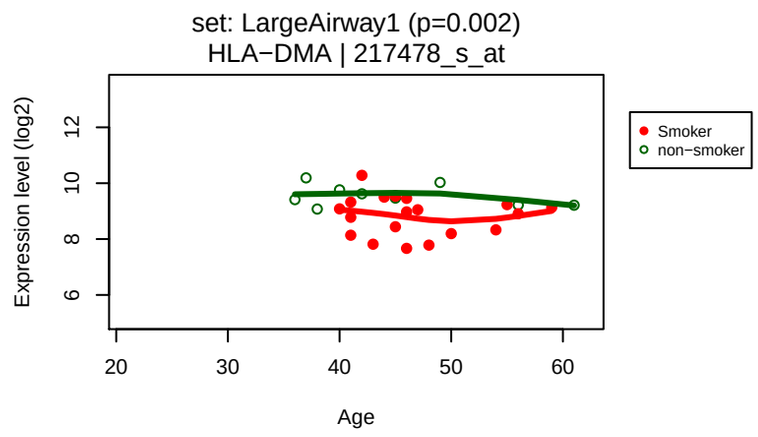

Gene expression of smokers and nonsmokers
gene: HLA−DMA | 217478_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4622
p=0.0112
p=0.0046
p=0.002
p=0.1374
p=0.9974
p=0.4313
p=0.4872
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

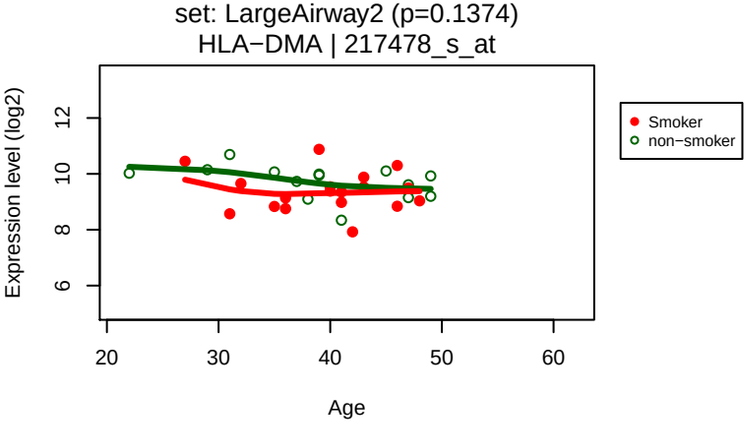

Gene expression ratio (smokers/nonsmokers)
gene: HLA−DMA | 217478_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4622
p=0.0112
p=0.0046
p=0.002
p=0.1374
p=0.9974
p=0.4313
p=0.4872
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

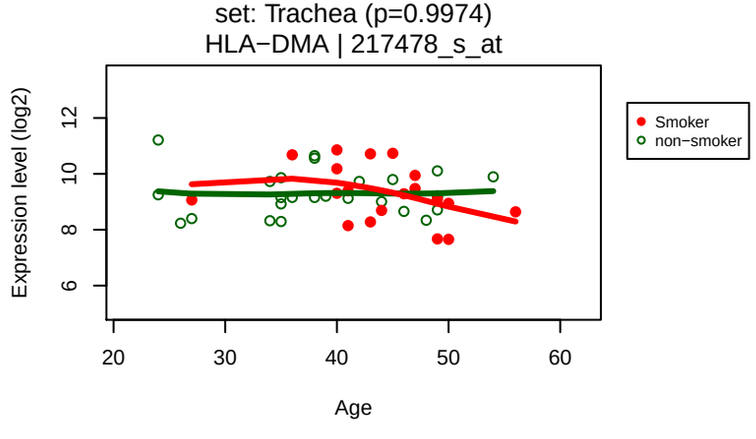

Gene expression ratio in Males
gene: HLA−DMA | 217478_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0037
p=0.1043
p=0.7838
p=0.3355
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−DMA | 217478_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8569
p=0.8144
p=0.8192
p=0.7358
SmallAirway2
LargeAirway2
Trachea
Macrophage2

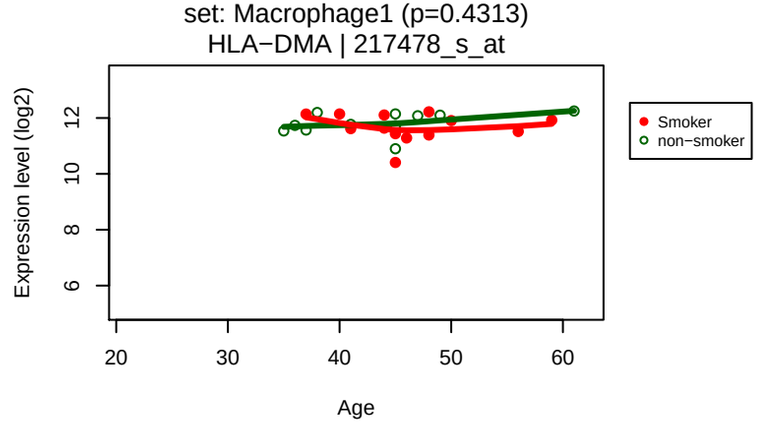

set: SmallAirway1 (p=0.4622)
HLA−DMA | 217478_s_at
Expression level (log2)
Age
Smoker
non−smoker

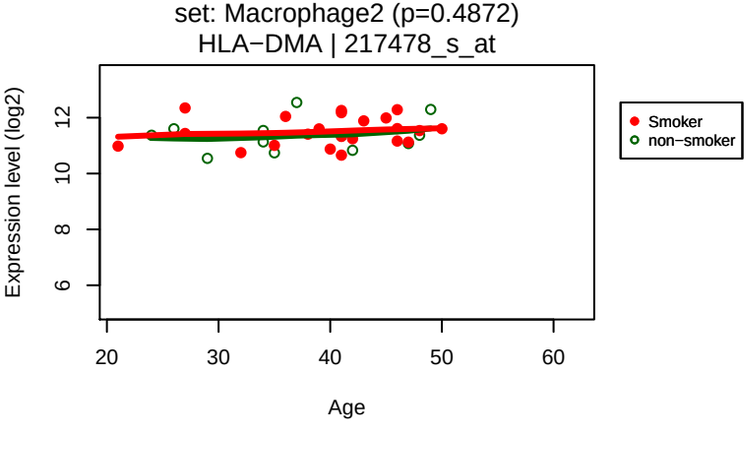

set: SmallAirway2 (p=0.0112)
HLA−DMA | 217478_s_at
Expression level (log2)
Age
Smoker
non−smoker

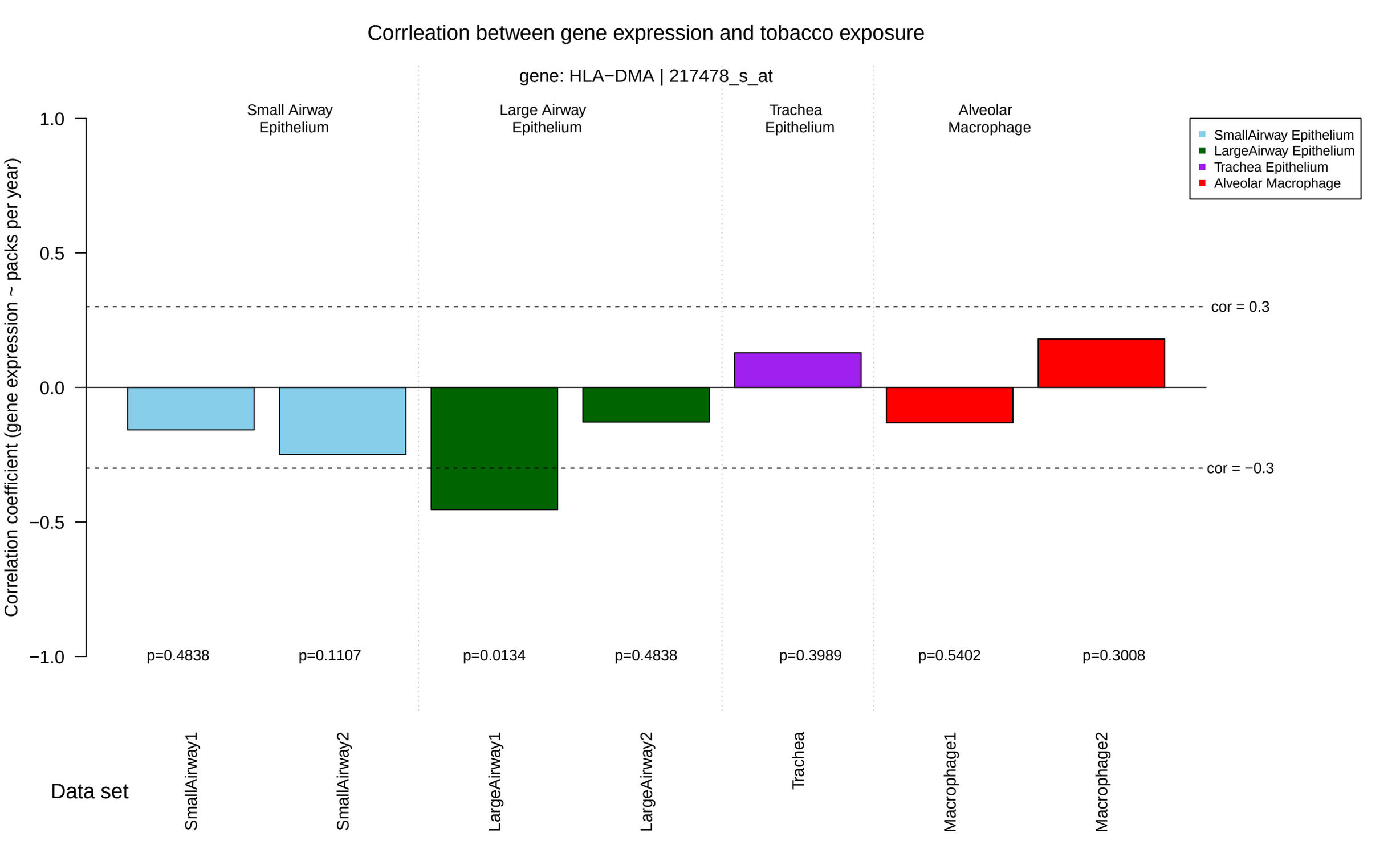

set: LargeAirway1 (p=0.002)
HLA−DMA | 217478_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1374)
HLA−DMA | 217478_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9974)
HLA−DMA | 217478_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4313)
HLA−DMA | 217478_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4872)
HLA−DMA | 217478_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−DMA | 217478_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4838
p=0.1107
p=0.0134
p=0.4838
p=0.3989
p=0.5402
p=0.3008
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2